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# Decoupling What from Where: A Mechanism Study of Action-Type Supervision for GUI Grounding in a Small Vision-Language Model

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## Abstract

1 A GUI agent decides which action to take and where on the screen to take it; native  
2 agent models emit both in one token stream. We ask what supervising the action  
3 type does for the grounding half of that decision, using Qwen2-VL-2B with LoRA  
4 on Android in the Wild. A flat baseline is compared with five ways of supplying  
5 the type under matched data, compute, adapters, and decoding: an auxiliary loss,  
6 a hard-routed action word, an additive learned embedding, a prepended learned  
7 token, and the type written into the prompt. With five seeds, a cluster bootstrap  
8 over validation examples, and seed-level paired tests, two results hold. On a train-  
9 ing stream that mixes clicks and scrolls with type events whose recorded target is  
10 a fixed off-screen point, the auxiliary loss, the additive embedding, and the type  
11 written into the prompt each raise click grounding by 6 to 10 points over the base-  
12 line; the prepended learned token, the mechanism we originally hypothesized, is  
13 no better than the baseline. The benefit is largely protection from the degenerate  
14 class: removing those events from training raises the baseline itself by four points,  
15 after which no mechanism beats it on the same validation examples, and on a  
16 separate stream of taps and swipes no mechanism helps either. Interventions on  
17 the trained models show that both learned embeddings are consulted at inference,  
18 with a difference that matters for deployment: with its embedding zeroed, the  
19 additive model returns to the baseline’s accuracy, whereas the prepended-token  
20 model falls below it. We also correct our own description of a silent failure: in-  
21 jecting the conditioning through `inputs_embeds` makes Qwen2-VL fall back to  
22 one-dimensional positions for image tokens, which produced a nine-point deficit  
23 that vanished once `input_ids` were preserved.

## 24 1 Introduction

25 A GUI agent maps a screenshot and an instruction to a low-level action: an action type such as *click*,  
26 *scroll*, or *type*, and, for spatial actions, a target location. Current native agent models (UI-TARS [Qin  
27 et al., 2025], OS-Atlas [Wu et al., 2025c], SeeClick [Cheng et al., 2024], ShowUI [Lin et al., 2025])  
28 produce both in a single autoregressive stream. The two decisions are different kinds of prediction:  
29 the action type is a small classification problem that depends on the goal and the screen state, and  
30 the location is a localization problem in which the model has to find the evidence on the screen that  
31 supports the action and commit to a point. Several recent agents separate the two, predicting the  
32 type first and the target conditioned on it [Ma et al., 2024, Christianos et al., 2025]. What has not  
33 been measured is what the type supervision itself does to the grounding model, and which of the  
34 many ways of delivering it matters.

This paper is a matched-compute study of that question in a small model. We fine-tune Qwen2-VL-2B [Wang et al., 2024] with LoRA [Hu et al., 2022] to emit a coordinate string, and compare a flat baseline with five conditioning mechanisms that receive the action type in different forms: as an auxiliary classification loss, as a hard-routed action word in the output, as an additive learned embedding, as a prepended learned token, and as a word in the prompt. Each variant sees the same data in the same order and uses the same optimizer, adapters, and decoding. We started from the hypothesis that the prepended token would be the right mechanism, because it gives the model a dedicated, attendable position for the action type.

The results support type supervision and refute our hypothesis about the mechanism, and they depend on a property of the data that we did not appreciate at first. Android in the Wild records type events with a touch point off the screen, which our coordinate serializer clamps to the origin, so a training stream that mixes clicks, scrolls, and type events contains a class whose target is a fixed point. On that stream, the auxiliary loss and the additive embedding raise hit@0.10 by +0.064 and +0.054 over the flat baseline (five seeds, cluster bootstrap over examples, seed-level  $p = 0.004$  and 0.028), almost entirely on clicks; the type written into the prompt as a word, with no new parameters, does as well, and the prepended token is within noise of the baseline. The obvious explanation, that the baseline learns to emit the off-screen point for clicks, accounts for little of the gap (it does so on 4% of clicks), but the class does hurt the baseline more broadly: removing type events from training while scoring the identical validation examples raises the baseline by four points, after which no mechanism beats it, and on a separate stream of taps and swipes none helps either.

Interventions on the trained models explain the ranking. A wrong type collapses either embedding model and a zeroed or class-averaged embedding costs accuracy in both, so both learned embeddings are consulted at inference; the difference is what is left when the signal is removed. The zeroed additive model sits at the flat baseline’s level, the zeroed prepended-token model seven points below it. An earlier implementation that injected the conditioning through `inputs_embeds`, which makes Qwen2-VL fall back to one-dimensional positions for its image tokens, scored nine points below the baseline, a result we initially read as evidence against conditioning.

Our contributions are (1) a matched-compute comparison of five action-type conditioning mechanisms against a flat baseline, with five seeds, a cluster bootstrap over examples, seed-level tests, and a control stream where conditioning should not help; (2) a diagnosis of where the gain comes from, including retraining with the degenerate class removed; (3) interventions that separate what each mechanism does at inference from what it does in training; and (4) a corrected account of a positional-encoding failure that inverted a conclusion.

## 2 Related Work

**GUI grounding and agent models.** Mind2Web [Deng et al., 2023] and Android in the Wild [Rawles et al., 2023] supply the web and mobile episodes we train on; AndroidControl [Li et al., 2024] and AndroidWorld [Rawles et al., 2025] extend mobile evaluation. SeeClick [Cheng et al., 2024] argued that grounding is the bottleneck for visual GUI agents and introduced ScreenSpot, which ScreenSpot-Pro [Li et al., 2025] extends. UGround [Gou et al., 2025] and OS-Atlas [Wu et al., 2025c] scale grounding pretraining; Aguvis [Xu et al., 2025], UI-TARS [Qin et al., 2025], CogAgent [Hong et al., 2024], ScreenAI [Baechler et al., 2024], and Ferret-UI [You et al., 2024b] emit action and location together, the design our flat baseline follows. GUI-Actor [Wu et al., 2025a] replaces coordinate text with an attention head, UI-R1 [Lu et al., 2026] rewards type and argument separately, RegionFocus [Luo et al., 2025] zooms at test time, and OmniParser [Lu et al., 2024] and Set-of-Mark [Yang et al., 2023] externalize grounding into a parser.

**Type-first agents.** CoCo-Agent [Ma et al., 2024] decomposes AITW action prediction into the type followed by the target conditioned on it, in one stream; that is our hard-routing variant. Li-MAC [Christianos et al., 2025] places a small action-type transformer ahead of a LoRA-tuned Qwen2-VL-2B on AITW and AndroidControl and selects click targets conditioned on the predicted type, using screen elements rather than pixels. SeeAct [Zheng et al., 2024], Ponder & Press [Wang et al., 2025], and Aria-UI [Yang et al., 2025] separate text planning from pixel grounding; GUI-Libra [Yang et al., 2026] reweights reasoning against action tokens and GUI-AIMA [Zhou et al., 2025] adds an anchor token for grounding. These systems establish that type-first pipelines work;

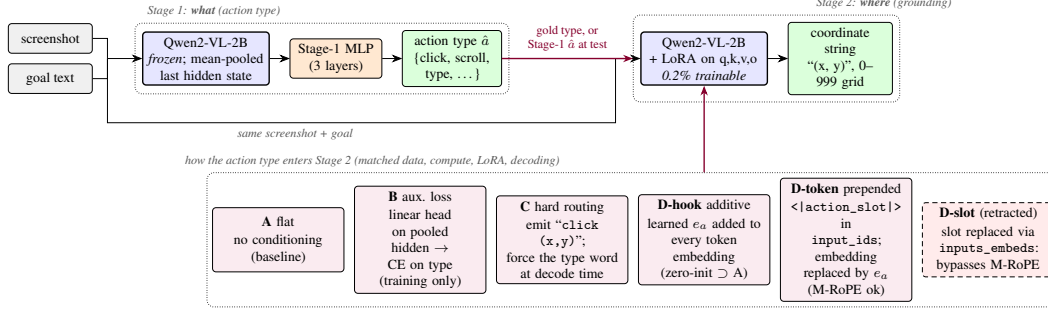


Figure 1: Two-stage pipeline. Stage 1 classifies the action type from mean-pooled frozen features. Stage 2 grounds the action to a coordinate string, and the action type enters it through one of the mechanisms in the bottom row (D-text, the type as a word in the prompt, is not drawn). All mechanisms share data, compute, adapters, and decoding. D-slot is the retracted variant whose `inputs_embeds` injection bypasses the position computation.

88 none holds data, compute, and decoding fixed across conditioning mechanisms or intervenes on the  
 89 trained model to ask whether the type signal is used, which is what this paper does.

90 **Conditioning, faithfulness, and the backbone.** SayCan [Ahn et al., 2022], ReAct [Yao et al.,  
 91 2023], and RT-2 [Brohan et al., 2023] are the reference points for factored, exposed, and token-level  
 92 action decisions; AutoUI [Zhang and Zhang, 2024], AppAgent [Zhang et al., 2025], DigiRL [Bai  
 93 et al., 2024], WebRL [Qi et al., 2025], and VSC-RL [Wu et al., 2025b] improve mobile agents by imi-  
 94 tation or reinforcement learning without changing how the type enters the grounding computation.  
 95 Our auxiliary-loss variant is a multi-task objective in the sense of Caruana [1997]. Kosmos-2 [Peng  
 96 et al., 2024], Shikra [Chen et al., 2023], and Ferret [You et al., 2024a] train VLMs to refer to image  
 97 regions, and POPE [Li et al., 2023] and HallusionBench [Guan et al., 2024] ask whether an output  
 98 is supported by the image; our attention measurements ask that of a coordinate. Qwen2-VL [Wang  
 99 et al., 2024] assigns three-dimensional rotary positions (extending RoPE [Su et al., 2024]) to visual  
 100 tokens from `input_ids`; Section 6.2 describes what happens when conditioning bypasses that path.

## 101 3 Method

### 102 3.1 Action taxonomy and grounding target

103 Both datasets are mapped onto a canonical eight-class action space (*click*, *double-click*, *type*, *scroll*,  
 104 *drag*, *hotkey*, *wait*, *finished*). Mind2Web’s CLICK, SELECT, and HOVER map to *click* and TYPE  
 105 to *type*. AITW stores gestures as touch and lift points; we split them into taps (*click*) and swipes  
 106 (*scroll*) by the normalized distance between the two, with threshold 0.04. The grounding target is  
 107 the touch point normalized to  $[0, 1]^2$ , serialized as  $(x, y)$  with integers on a 0–999 grid so every  
 108 target has the same token length. AITW records non-touch actions such as typing with the touch  
 109 point  $(-1, -1)$ ; the serializer clamps this to  $(0, 0)$ , so every *type* event in the training stream has  
 110 the same target at the screen origin, and its distance to the raw target is exactly  $\sqrt{2}$  whatever the  
 111 model predicts. This detail matters for the analysis in Section 5.1.

### 112 3.2 Stage 1: which action

113 Stage 1 runs the frozen Qwen2-VL-2B-Instruct backbone over the screenshot and instruction, mean-  
 114 pools the final hidden states over unmasked tokens into a 1536-dimensional vector, and trains a three-  
 115 layer MLP ( $1536 \rightarrow 1024 \rightarrow 256 \rightarrow 8$ , dropout 0.1) with class-weighted cross-entropy, AdamW  
 116 (learning rate  $10^{-4}$ , weight decay 0.01), batch size 128, for eight epochs, keeping the checkpoint  
 117 with the best validation macro-F1. We compare features from text only, text plus screenshot, and a  
 118 sanity variant with zeroed features.

### 119 3.3 Stage 2: where

120 Stage 2 fine-tunes the same backbone with LoRA adapters (rank 16,  $\alpha = 32$ , dropout 0.05) on the  
 121 query, key, value, and output projections, 4.4M trainable parameters out of 2.2B. The prompt is the  
 122 screenshot, `Goal: {goal}`, and a fixed instruction to predict the action coordinate; the answer is  
 123 the coordinate string. With  $s$  the screenshot,  $g$  the goal,  $a$  the action type, and  $y_{1:T}$  the coordinate  
 124 tokens, every variant minimizes

$$\mathcal{L}_{\text{coord}} = - \sum_{t=1}^T \log p_{\theta}(y_t | y_{<t}, s, g, c), \quad (1)$$

125 masked to the answer tokens, with  $c$  the variant-specific conditioning.

126 **A, flat.**  $c = \emptyset$ : the joint-decoding design of native agent models, reduced to the grounding sub-  
 127 problem.

128 **B, auxiliary loss.** A linear head  $h_{\phi}$  on the mean-pooled last hidden state  $\bar{z}$  predicts the action type  
 129 during training,  $\mathcal{L}_B = \mathcal{L}_{\text{coord}} + \lambda \text{CE}(h_{\phi}(\bar{z}), a)$  with  $\lambda = 1$ . Nothing about the type is provided at  
 130 inference, so any gain is a training effect.

131 **C, hard routing.** The model is trained to emit the action word before the coordinate, `click (x,`  
 132 `y)`, as in CoCo-Agent’s conditional action prediction. At test time the word for the gold (or Stage-1)  
 133 type is forced as a decoding prefix.

134 **D-hook, additive embedding.** A learned table  $E \in \mathbb{R}^{8 \times d}$  holds one vector per class; a forward hook  
 135 on the embedding layer adds  $E[a]$  to every text and image token embedding.  $E$  is zero-initialized,  
 136 so D-hook starts as an exact copy of A.

137 **D-token, prepended action token.** A new token `<|action_slot|>` is added to the vocabulary  
 138 and placed at the start of the text prompt, so it occupies a real position in `input_ids`; a forward  
 139 hook replaces its embedding with  $E[a]$ , initialized from  $\mathcal{N}(0, 0.02^2)$ . This is the architecture we  
 140 originally hypothesized: a full-norm, attendable representation of the type.

141 **D-text, type as a word.** The prompt begins with `Action: click.` (or the gold type’s word),  
 142 using the pretrained word embeddings and no new parameters. This is the obvious baseline for the  
 143 two learned-embedding variants.

144 For C, D-hook, D-token, and D-text the gold type is used in training and at evaluation unless stated  
 145 otherwise; Section 5.3 closes the loop with Stage-1 predictions. All variants share the data order, the  
 146 optimizer (AdamW, learning rate  $2 \times 10^{-5}$ , weight decay 0.01, batch size 1, gradient clipping at 1.0,  
 147 two epochs), and greedy decoding with 16 new tokens. No hyperparameter was tuned per variant;  
 148 Section 7 returns to this.

## 149 4 Experimental setup

150 **Data.** Stage 1 uses 7,775 Multimodal-Mind2Web training steps evaluated on the 1,338-step  
 151 `test_task` split, and 5,000 AITW steps from the General subset evaluated on 1,000 held-out  
 152 steps. Stage 2 uses two AITW streams built by filtering the training split in its stored order.  
 153 `all_with_coords` keeps taps, swipes, and type events; at 1,200 training examples its validation  
 154 slice of 250 steps holds 169 clicks, 46 scrolls, and 35 type events. `taps_and_swipes` keeps only  
 155 taps and swipes (1,000 training and 200 validation steps); both land anywhere on the screen. The  
 156 AITW stream is deterministic, so for a given mix and training size every seed and every variant  
 157 is evaluated on the same examples in the same order, which is what makes paired tests possible.  
 158 The validation slice begins where the training slice ends; consecutive steps come from the same  
 159 episodes, and absolute numbers are not comparable across training sizes. A second control uses  
 160 Multimodal-Mind2Web (1,000 training and 250 evaluation steps), where the target is the center of  
 161 the gold element. The Stage-2 study is run on AITW because that is where the action type is a vision  
 162 problem: Stage 1 (Table 5 in the appendix) gains +0.414 macro-F1 from the screenshot on AITW,  
 163 whose instructions are goals, and +0.009 on Mind2Web, whose instructions name the action.

164 **Metrics.** `hit@r` is the fraction of predictions within normalized Euclidean distance  $r$  of the target,  
 165 for  $r \in \{0.05, 0.10, 0.25\}$ ; `hit@0.10` is the headline. We also report the mean normalized distance.

Table 1: Stage 2 grounding on AITW all\_with\_coords ( $n_{\text{train}}=1200$ ,  $n_{\text{val}}=250$ , 5 seeds). Mean  $\pm$  std over seeds;  $\Delta$  with 95% paired-bootstrap CI over pooled (seed, example) units; \* $p < .05$ , \*\* $p < .01$ , \*\*\* $p < .001$ .

| Variant                | hit@0.05          | hit@0.10          | hit@0.25          | mean L2 $\downarrow$ | $\Delta$ hit@0.10 vs. A [cluster CI] | seed-level $p$ |
|------------------------|-------------------|-------------------|-------------------|----------------------|--------------------------------------|----------------|
| A: flat                | 0.130 $\pm$ 0.039 | 0.229 $\pm$ 0.043 | 0.499 $\pm$ 0.029 | 0.406 $\pm$ 0.028    |                                      |                |
| B: aux. loss           | 0.178 $\pm$ 0.025 | 0.293 $\pm$ 0.021 | 0.582 $\pm$ 0.026 | 0.365 $\pm$ 0.004    | +0.064 [+0.036, +0.093]***           | 0.004          |
| C: hard routing        | 0.170 $\pm$ 0.022 | 0.264 $\pm$ 0.045 | 0.538 $\pm$ 0.059 | 0.421 $\pm$ 0.016    | +0.035 [+0.001, +0.069]*             | 0.189          |
| D-hook: additive       | 0.170 $\pm$ 0.042 | 0.282 $\pm$ 0.041 | 0.563 $\pm$ 0.016 | 0.370 $\pm$ 0.010    | +0.054 [+0.022, +0.086]***           | 0.028          |
| D-token: prepended     | 0.144 $\pm$ 0.056 | 0.238 $\pm$ 0.049 | 0.504 $\pm$ 0.055 | 0.386 $\pm$ 0.021    | +0.009 [-0.015, +0.034]              | 0.465          |
| D-text: word in prompt | 0.186 $\pm$ 0.022 | 0.302 $\pm$ 0.028 | 0.568 $\pm$ 0.033 | 0.364 $\pm$ 0.008    | +0.073 [+0.033, +0.113]***           | 0.018          |

Table 2: Where the gain lives. Per-class hit@0.10 on the headline mix (five seeds, same basis as Table 1), the fraction of click predictions that are the clamped type target (0, 0), and click hit@0.10 after excluding those predictions. Every model scores zero on *type* because the raw target is off-screen.

| Variant            | click             | scroll            | type  | clicks sent to (0, 0)          | click, excluding those |
|--------------------|-------------------|-------------------|-------|--------------------------------|------------------------|
| A: flat            | 0.256 $\pm$ 0.062 | 0.304 $\pm$ 0.034 | 0.000 | 4.3% (13.0% on the worst seed) | 0.266                  |
| B: aux. loss       | 0.357 $\pm$ 0.026 | 0.278 $\pm$ 0.018 | 0.000 | 2.4%                           | 0.366                  |
| C: hard routing    | 0.350 $\pm$ 0.061 | 0.148 $\pm$ 0.073 | 0.000 | 0.0%                           | 0.350                  |
| D-hook: additive   | 0.321 $\pm$ 0.064 | 0.357 $\pm$ 0.039 | 0.000 | 0.0%                           | 0.321                  |
| D-token: prepended | 0.269 $\pm$ 0.069 | 0.304 $\pm$ 0.090 | 0.000 | 0.0%                           | 0.269                  |

Unparseable outputs count as misses at distance  $\sqrt{2}$ ; only hard routing produces them (2 to 8% of outputs), and its numbers are reported both ways. Per-class numbers are computed on the same basis as the overall ones.

**Statistics.** The headline setting uses five seeds (42 to 46); every other cell uses three (42 to 44). We report the mean and sample standard deviation over seeds. Every run logs a per-example distance on a shared evaluation set, and every seed of every variant shares the per-seed data order, so comparisons are paired at two levels. Our primary interval is a cluster bootstrap that resamples validation examples with replacement and keeps all seeds of a sampled example together (10,000 resamples, 95% percentile interval, two-sided bootstrap  $p$  [Efron and Tibshirani, 1993]); resampling (seed, example) units separately would treat the seeds of one example as independent and gives intervals about a third narrower. We also report a paired  $t$ -test on the five (or three) per-seed means, which has little power but cannot be fooled by within-seed correlation, and treat a difference as established when both agree; stars come from the cluster bootstrap with the seed-level  $p$  beside them. No correction is made for the number of contrasts; the scaling study in Section 5.4 is presented as descriptive for that reason.

## 5 Results

### 5.1 Type supervision improves click grounding on the mixed stream

Table 1 is the main result: six variants on all\_with\_coords at 1,200 training examples, five seeds each. The auxiliary loss (B) and the additive embedding (D-hook) beat the flat baseline on every metric, by +0.064 and +0.054 hit@0.10 with cluster intervals that exclude zero and seed-level  $p$  of 0.004 and 0.028, and they are indistinguishable from each other (D-hook – B:  $-0.010$ , cluster CI  $[-0.043, +0.024]$ , seed  $p = 0.56$ ). Hard routing (C) gains +0.035 with a cluster interval that barely excludes zero and a seed-level  $p$  of 0.19, and loses on mean distance, partly because forcing the action word makes 2 to 8% of its outputs unparseable; scored on parsed outputs only, its click rate is unchanged (0.356 vs. 0.350). The prepended token (D-token), our hypothesized mechanism, is within noise of the baseline on every hit rate (+0.009, cluster CI  $[-0.015, +0.034]$ ) and below B by  $-0.055$  (cluster CI  $[-0.088, -0.024]$ , seed  $p = 0.016$ ). The type written into the prompt as a word (D-text), which adds no parameters, gains +0.057 over the baseline (cluster CI  $[+0.031, +0.084]$ , seed  $p = 0.04$ ) and is within noise of B (+0.007, cluster CI  $[-0.020, +0.035]$ ), so the failure is specific to the learned prepended token, not to conditioning at the input. Seed variance is large: the baseline ranges from 0.169 to 0.280 across seeds, which is why we report five seeds and paired statistics rather than a best run; the ordering with the first three seeds was the same.

Table 3: Control: AITW `taps_and_swipes` ( $n_{\text{train}}=1000$ ,  $n_{\text{val}}=200$ , 3 seeds), where action type is not spatially informative.

| Variant          | hit@0.05          | hit@0.10          | hit@0.25          | mean L2 ↓         | $\Delta\text{hit@0.10 vs. A}$ [cluster CI] | seed-level $p$ |
|------------------|-------------------|-------------------|-------------------|-------------------|--------------------------------------------|----------------|
| A: flat          | 0.243 $\pm$ 0.020 | 0.382 $\pm$ 0.015 | 0.720 $\pm$ 0.017 | 0.185 $\pm$ 0.003 |                                            |                |
| B: aux. loss     | 0.225 $\pm$ 0.015 | 0.368 $\pm$ 0.012 | 0.730 $\pm$ 0.039 | 0.173 $\pm$ 0.011 | -0.013 [-0.038, +0.012]                    | 0.456          |
| C: hard routing  | 0.217 $\pm$ 0.016 | 0.325 $\pm$ 0.040 | 0.635 $\pm$ 0.064 | 0.271 $\pm$ 0.068 | -0.057 [-0.108, -0.007]*                   | 0.215          |
| D-hook: additive | 0.235 $\pm$ 0.022 | 0.363 $\pm$ 0.021 | 0.725 $\pm$ 0.043 | 0.182 $\pm$ 0.009 | -0.018 [-0.052, +0.015]                    | 0.334          |

Table 2 shows where the gain lives and tests the simplest explanation for it. The improvement is a click improvement: B and C gain ten points on clicks, D-hook six, and C pays for its click gain with a sixteen-point loss on scrolls because forcing the word *scroll* into the output derails the gesture; D-hook is the only variant that improves both classes. Since the only difference between this stream and the control is the type class with its fixed target, one might expect the flat baseline’s click errors to be emissions of that target that the conditioned models avoid. They mostly are not. The baseline emits (0, 0) for 4.3% of clicks (13% on its worst seed), the auxiliary-loss model for 2.4%, and the three conditioned-at-inference models never; excluding those predictions moves the baseline’s click rate from 0.256 to 0.266 while B stays at 0.366 and D-hook at 0.321. The class degrades the baseline’s click grounding more broadly than by exact emissions, and Table 7 in the appendix measures how much: we retrain A, B, and D-hook on the same training slice with the type events removed and score the identical 250 validation examples (three seeds). Removing the class raises the flat baseline from 0.255 to 0.297 hit@0.10 (cluster CI on the change [+0.016, +0.069]; clicks from 0.294 to 0.365) and leaves the conditioned models where they were. On that type-free stream the additive embedding’s advantage over the stronger baseline vanishes (+0.008, cluster CI [-0.020, +0.035]; on clicks -0.008), and the auxiliary loss keeps only a small click advantage (+0.039, cluster CI [+0.004, +0.075]) and a hit@0.25 advantage (+0.053, CI [+0.024, +0.083]) with no overall hit@0.10 gain (+0.021, CI [-0.004, +0.045]). With the control in the next section, we read this as follows: what type supervision buys on the mixed stream is protection of click grounding from a class whose target is degenerate, and we do not claim a general spatial prior.

## 5.2 Control: no gain on taps and swipes

On `taps_and_swipes` (Table 3), whose training stream has no degenerate class, B and D-hook are within noise of the baseline (-0.013 and -0.018 hit@0.10, cluster intervals [-0.038, +0.012] and [-0.052, +0.015]), so the data admit harms of up to four points as well as no effect, and C is worse by -0.057 (cluster CI [-0.108, -0.007]) with a large increase in mean distance. The Multimodal-Mind2Web control sits at floor (a tenth of predicted points fall inside the gold element for both A and D-hook) and is uninformative; it is reported in the appendix and not leaned on.

## 5.3 End to end: predicted types instead of gold

The tables above condition on the gold action type. To close the loop we train D-hook, fit the Stage-1 MLP on frozen features of its training examples for a fixed 200 full-batch epochs with no checkpoint selection, and evaluate D-hook on the same validation examples with gold types and with Stage-1 predictions (three seeds; Stage 1 reaches 0.843 accuracy). The oracle pipeline scores 0.292 hit@0.10, the predicted-type pipeline 0.271, and the flat baseline on the same seeds 0.255. Only the oracle-to-predicted gap is established (+0.021, cluster CI [+0.003, +0.041]); the predicted pipeline’s margin over the baseline (+0.016, cluster CI [-0.013, +0.047], seed  $p = 0.44$ ) is not, and we report it as descriptive. Classifier error removes more than half of the conditioning gain in this setting.

## 5.4 Scaling: a descriptive study

Figure 2 extends the comparison to  $n \in \{300, 500, 800, 1200, 2500, 5000\}$  with three seeds at every size. The cluster intervals for D-hook - A exclude zero at five of the six sizes on hit@0.10 (all six on hit@0.25), and its estimate at 5,000 examples (+0.043) is the same size as at 1,200; B - A excludes zero at 300 and 1,200, is negative at 800, and is within noise elsewhere. With three seeds the seed-level tests reach  $p < 0.05$  only at 300 and 5,000 for D-hook and at 300 and 1,200 for B, and a direct D-hook - B contrast is within noise at four of the six sizes. We read the study as consistent with the additive embedding’s advantage persisting with data and the auxiliary loss’s not, and no

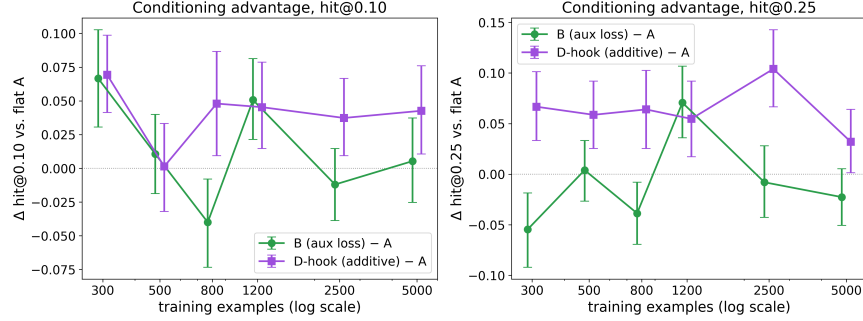


Figure 2: Conditioning advantage over the flat baseline as a function of training-set size on all\_with\_coords, three seeds per cell, 95% cluster-bootstrap intervals over examples. Each size is scored on a different validation slice with a different class mix (Table 8), so only within-size differences are meaningful, and 24 contrasts are shown without correction.

Table 4: Interventions on the trained conditioned models: the same model is decoded with the gold action embedding, a wrong one (cyclic permutation; click→type), a wrong one that swaps only clicks and scrolls, the table zeroed, and every row replaced by the class mean (uninformative, in-distribution norm). Five seeds × 250 examples;  $\Delta$  = gold – condition with 95% cluster-bootstrap CI over examples and the seed-level paired  $p$ . Flat A on the same examples is the reference row.

| Model              | Conditioning          | hit@0.10      | click hit@0.10 | scroll hit@0.10 | $\Delta$ gold – cond. (hit@0.10) | seed $p$ |
|--------------------|-----------------------|---------------|----------------|-----------------|----------------------------------|----------|
| D-hook: additive   | gold type             | 0.275 ± 0.040 | 0.310 ± 0.060  | 0.357 ± 0.052   |                                  |          |
|                    | wrong (cyclic)        | 0.054 ± 0.009 | 0.006 ± 0.000  | 0.270 ± 0.050   | +0.222 [+0.174, +0.270]***       | 0.000    |
|                    | wrong (click↔scroll)  | 0.099 ± 0.056 | 0.073 ± 0.087  | 0.270 ± 0.050   | +0.176 [+0.133, +0.221]***       | 0.003    |
|                    | zeroed                | 0.248 ± 0.032 | 0.288 ± 0.053  | 0.291 ± 0.050   | +0.027 [-0.006, +0.060]          | 0.224    |
|                    | class mean            | 0.206 ± 0.058 | 0.217 ± 0.099  | 0.322 ± 0.064   | +0.070 [+0.034, +0.106]***       | 0.133    |
|                    | flat A, same examples | 0.229 ± 0.043 | 0.256          | 0.304           | +0.046 [+0.017, +0.077]**        | 0.032    |
| D-token: prepended | gold type             | 0.242 ± 0.054 | 0.272 ± 0.079  | 0.313 ± 0.099   |                                  |          |
|                    | wrong (cyclic)        | 0.079 ± 0.012 | 0.036 ± 0.019  | 0.300 ± 0.054   | +0.162 [+0.120, +0.205]***       | 0.001    |
|                    | wrong (click↔scroll)  | 0.155 ± 0.043 | 0.148 ± 0.063  | 0.300 ± 0.054   | +0.086 [+0.043, +0.130]***       | 0.073    |
|                    | zeroed                | 0.162 ± 0.072 | 0.160 ± 0.112  | 0.291 ± 0.033   | +0.080 [+0.050, +0.112]***       | 0.120    |
|                    | class mean            | 0.169 ± 0.043 | 0.169 ± 0.067  | 0.296 ± 0.025   | +0.073 [+0.040, +0.107]***       | 0.091    |
|                    | flat A, same examples | 0.229 ± 0.043 | 0.256          | 0.304           | +0.013 [-0.010, +0.037]          | 0.374    |

stronger than that; an earlier single-seed curve had suggested that all conditioning gains shrink with data, which three seeds do not support. Figure 4 in the appendix shows individual screens.

## 6 Mechanism

### 6.1 Interventions on both embedding mechanisms

D-token’s failure to help could mean that the model ignores the slot, and D-hook’s success could be a training effect that leaves a vector the model never reads. To test both on equal terms we retrain each at the headline setting (five seeds) and decode every trained model under five conditionings on the full validation slice: the gold type; a wrong type from a cyclic permutation of the classes present (click to type, type to scroll, scroll to click); a second wrong map that swaps only clicks and scrolls, so no groundable example is routed to the degenerate class; the table zeroed; and every row replaced by the mean of the rows of the classes present, which is uninformative but in-distribution. Table 4 reports the results with the flat baseline on the same examples; an earlier three-condition test on separately trained D-token models (Table 9, appendix) agrees. The learned rows of D-hook are small: each has norm 0.04 against a mean token-embedding norm of 0.58, whereas D-token’s rows have norm 0.78. Small as they are, the model reads them. A wrong type collapses D-hook from 0.275 to 0.054 under the cyclic map and to 0.099 under the click-scroll map (cluster intervals [+0.174, +0.270] and [+0.133, +0.221], seed  $p < 0.001$  and 0.003), so the collapse is not an artifact of routing clicks to the degenerate class. The class-mean vector costs seven points (+0.070, cluster CI [+0.034, +0.106], seed  $p = 0.13$ ). Zeroing costs about three, and that difference is not established (+0.027, cluster CI [-0.006, +0.060], seed  $p = 0.22$ ): the zeroed model sits at the level of the flat baseline (+0.019 over A, not established). The single-seed attention run in Section ??, on

263 which an earlier version of this paper based the claim that the embedding could be removed without  
264 loss, is consistent with these intervals and was simply underpowered. The additive embedding  
265 therefore behaves as a switch at inference: with the right type the model gains six points over the  
266 baseline, without a signal it returns to the baseline’s level, and with a wrong or blended signal it  
267 collapses.

268 D-token differs in one respect that matters for deployment. Every perturbation collapses it, including  
269 the in-distribution class-mean condition ( $+0.073$ , cluster CI  $[+0.040, +0.107]$ ), so its sensitivity is  
270 not an artifact of an out-of-distribution zero vector, and the zeroed model is worse than the flat  
271 baseline by  $-0.067$  (cluster CI  $[-0.098, -0.038]$ ), whereas the zeroed D-hook is not. Adaptation  
272 with a prepended token makes the model depend on the token; adaptation with an additive vector  
273 does not, even though the vector is used. In both mechanisms the conditioning acts almost entirely  
274 on clicks. Attention measured at the token that predicts the y coordinate moves the same way as  
275 accuracy under these interventions on a single seed (Appendix G); we treat those numbers as a  
276 consistency check, not as evidence about faithfulness on their own.

## 277 6.2 A positional-encoding failure that inverted a conclusion

278 Our first implementation of the prepended token replaced the slot’s embedding by building  
279 `inputs_embeds` and passing that to the model instead of `input_ids`. On `taps_and_swipes` that  
280 variant scored  $0.298 \pm 0.026$  hit@0.10 against  $0.390 \pm 0.010$  for the flat baseline in the same set  
281 of runs, and a variant with the same slot but a frozen random embedding scored the same (0.290),  
282 which pointed at the injection path. Qwen2-VL derives its three-dimensional rotary positions from  
283 `input_ids`; given `inputs_embeds` alone, transformers falls back to one-dimensional positions  
284 replicated across the three axes, and the image tokens lose their two-dimensional layout (details, ver-  
285 sion, and a related library report [Hugging Face Transformers contributors, 2024] in Appendix F).  
286 Modifying embeddings through a forward hook with `input_ids` intact removed the deficit. We  
287 record this because the failure is silent and we initially read it as evidence against conditioning.

## 288 7 Limitations

289 The study is small: one 2B backbone, LoRA adapters, validation slices of 200 to 250 steps, one  
290 learning rate and one adapter configuration for every variant, and one L4 per run. The absolute  
291 grounding numbers are far below the field’s leaderboards, per-seed variance is large, and the conclu-  
292 sions rest on paired statistics reported at two levels because neither is clean: the cluster bootstrap  
293 ignores that seeds share a data order, and the seed-level test has five or three units. The refutation  
294 of the prepended token is conditional on that untuned configuration; a stronger D-token could ex-  
295 ist under a schedule that trains the fresh token faster, and we did not train a control that keeps the  
296 slot but freezes its embedding. The headline stream’s only spatially informative class has a fixed  
297 off-screen target, so the claim is about adaptation on a mixed stream that contains such a class, not  
298 about a general spatial prior. The end-to-end result is descriptive, the scaling study scores each size  
299 on a different validation slice with 24 uncorrected contrasts, the attention analysis uses one seed  
300 and one layer with a probe token chosen after the fact, and Mind2Web is at floor. We do not eval-  
301 uate on ScreenSpot or on multi-step task success, where the compounding-error motivation of the  
302 introduction would have to be tested directly. Code, split indices, and prompts will be released.

## 303 8 Conclusion

304 Supervising the action type during adaptation of a small grounding model helps when the training  
305 stream mixes groundable actions with a class whose target is degenerate; the help is concentrated on  
306 clicks, it is protection from that class rather than a spatial prior, and without such a class conditioning  
307 does nothing. An auxiliary loss, an additive embedding, and the type written into the prompt capture  
308 the gain; the prepended learned token we set out to validate is consulted by the model and improves  
309 nothing, and unlike the additive embedding it leaves a model that grounds below the baseline when  
310 the signal is removed. The lessons for adapting small VLMs on mixed action streams are to supervise  
311 the action type, to prefer conditioning that degrades to the baseline rather than below it, and to keep  
312 `input_ids` on the path whenever conditioning touches the embeddings.

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## A Training and evaluation details

**Data.** AITW screenshots in the mirror we use are stored as raw RGB byte buffers without an image header; we decode them by matching the buffer length against seven observed resolutions (for example  $540 \times 1140$  and  $412 \times 732$ ) with a portrait-aspect fallback. Mind2Web screenshots are downsampled so that no image exceeds one megapixel. Examples are drawn from the AITW training stream in stored order, filtered to the mix’s action labels; the first  $n$  eligible steps form the training slice and the next 250 (200 for the control mix) form validation. Consecutive steps belong to the same episodes, so a validation slice spans a few dozen episodes and may share its boundary episode with training. Table 8 lists the class mix of each validation slice.

**Prompts.** Stage 2 prompts contain the screenshot, the line `Goal: {goal}`, and the instruction `Predict the action coordinate.` (C uses `Predict the action type and coordinate.` and is trained to answer with the action word before the coordinate; D-text prefixes `Action: {word}.`). D-token places `<|action_slot|>` at the start of the text; exactly one slot per sequence is asserted at every step.

**Optimization.** AdamW with learning rate  $2 \times 10^{-5}$  and weight decay 0.01, batch size 1, gradient clipping at 1.0, two epochs, a fixed per-epoch shuffle seeded by the run seed. The action-embedding table (D-hook, D-token) and the auxiliary head (B) share the base learning rate. Evaluation is greedy with at most 16 new tokens; the coordinate is parsed with a regular expression and an unparseable output counts as a miss at distance  $\sqrt{2}$ . Greedy evaluation is deterministic for a trained model; models in different tables that share a seed are separately trained copies (Table 9 and the intervention runs retrain the variant), and their numbers differ by training nondeterminism.

**Compute.** Everything runs on single NVIDIA L4 GPUs. A Stage-2 run at 1,200 training examples takes about 35 minutes; the full study, including retracted and re-run cells, cost about 130 GPU-hours.

**Statistics.** For each contrast we align the per-example distances of the two variants on every shared seed. The cluster bootstrap resamples the validation examples with replacement and keeps all seeds of a sampled example together, recomputes the mean difference 10,000 times, and reports the percentile interval and the two-sided bootstrap  $p$ . The seed-level test is a paired  $t$ -test on the per-seed means. The earlier (seed, example)-unit bootstrap and its permutation counterpart are reported in the released logs; their intervals are narrower by about a third and every star in this paper was recomputed on the cluster basis.

## B Stage 1, Mind2Web, and per-class tables

Table 5: Stage 1 action-type macro-F1 (three seeds). On Mind2Web the instruction names the action, so the screenshot adds little; on AITW the instruction describes a goal and the screenshot is what separates the classes.

| Features                     | Mind2Web (2 classes) | AITW (5 classes)  |
|------------------------------|----------------------|-------------------|
| text only                    | $0.597 \pm 0.011$    | $0.141 \pm 0.032$ |
| text + screenshot            | $0.605 \pm 0.016$    | $0.555 \pm 0.036$ |
| zeroed (sanity)              | $0.469 \pm 0.000$    | $0.110 \pm 0.051$ |
| TF-IDF + logistic regression | 0.622                | 0.168             |
| majority class               | 0.472                | 0.140             |

Table 6: Multimodal-Mind2Web grounding control (1,000 training and 250 evaluation steps, three seeds, seed-level only). The target is the center of the gold element; hit@bbox is the fraction of predicted points inside the element. Both models are at floor.

| Variant          | hit@0.10          | hit@0.25          | hit@bbox          | mean L2 ↓         |
|------------------|-------------------|-------------------|-------------------|-------------------|
| A: flat          | $0.365 \pm 0.031$ | $0.645 \pm 0.035$ | $0.095 \pm 0.018$ | $0.234 \pm 0.023$ |
| D-hook: additive | $0.355 \pm 0.032$ | $0.664 \pm 0.052$ | $0.104 \pm 0.034$ | $0.233 \pm 0.027$ |

Table 7: Removing *type* events from the training slice only, scored on the identical headline validation slice (three seeds). Deltas are paired cluster bootstraps over examples; seed-level  $p$  in parentheses.

| Variant          | Training stream | hit@0.10          | click hit@0.10    | scroll hit@0.10   | $\Delta$ hit@0.10 vs. A (same stream) |
|------------------|-----------------|-------------------|-------------------|-------------------|---------------------------------------|
| A: flat          | with type       | $0.255 \pm 0.026$ | $0.294 \pm 0.042$ | $0.304 \pm 0.022$ |                                       |
| A: flat          | type removed    | $0.297 \pm 0.024$ | $0.365 \pm 0.039$ | $0.275 \pm 0.013$ |                                       |
| B: aux. loss     | with type       | $0.305 \pm 0.015$ | $0.373 \pm 0.020$ | $0.290 \pm 0.013$ |                                       |
| B: aux. loss     | type removed    | $0.319 \pm 0.034$ | $0.404 \pm 0.063$ | $0.246 \pm 0.066$ | +0.021 [-0.004, +0.045] (0.391)       |
| D-hook: additive | with type       | $0.300 \pm 0.035$ | $0.351 \pm 0.048$ | $0.341 \pm 0.045$ |                                       |
| D-hook: additive | type removed    | $0.305 \pm 0.050$ | $0.357 \pm 0.074$ | $0.348 \pm 0.000$ | +0.008 [-0.020, +0.035] (0.858)       |

## C Type events removed from training

## D Full scaling table

## E The earlier three-condition D-token test

Table 9 is the test that preceded Table 4: separately trained D-token models (five seeds), decoded with the gold embedding, the cyclic wrong map, and the table zeroed. Its numbers differ from the D-token rows of Table 4 by training nondeterminism and agree in every conclusion.

## F The inputs\_embeds failure in detail

Qwen2-VL locates the image tokens in `input_ids` to assign each visual token a temporal, height, and width position for its three-dimensional rotary embedding. When a caller supplies `inputs_embeds` and no `input_ids`, the model cannot perform that lookup; in the transformers releases our runs used (pinned at 4.45 or later; the current 5.9 release behaves the same) the positions become a one-dimensional sequence index replicated across the three axes, so every image token is positioned as text and the two-dimensional layout the grounding task depends on is lost. A caller on that path must also merge the visual features into the embeddings itself, a second opportunity for silent error. We inferred the cause from the code and from the frozen-embedding control described in Section 6.2 rather than by dumping position ids at the time. The same dependence on `input_ids` was reported as a generation crash in December 2024 [Hugging Face Transformers contributors, 2024]; the silent degradation when images are present is the case a fine-tuning practitioner meets. In the set of runs where the failure appeared, the fixed additive variant scored  $0.392 \pm 0.043$  hit@0.10 on `taps_and_swipes` against the flat baseline’s  $0.390 \pm 0.010$ ; the re-run of that control with per-example logging (Table 3) gives  $0.363 \pm 0.021$  against  $0.382 \pm 0.015$ .

## G Where the model looks

We measured where the models look, with the caveats that the measure is head-averaged attention at one layer, that attention is not information flow, and that teacher-forcing the gold answer places the gold  $x$  coordinate in the prefix of the token that predicts  $y$ . Under gold conditioning both embedding models put 10 to 11% of that token’s image attention within 0.10 of the target, against a chance share of 2.6%; the last prompt token, which our earlier visualizations used, puts about 4% there under every conditioning, so most of the localization we can see happens after the  $x$  coordinate is in the prefix. Teacher-forcing the model’s own answer instead of the gold one, so that the  $y$ -predicting token sees the predicted  $x$ , lowers the target share only slightly (D-hook 0.092, D-token 0.085), so

Table 8: Conditioning advantage vs. training-set size (AITW all\_with\_coords, 3 seeds per cell). Intervals are 95% cluster bootstraps over validation examples (all seeds of a sampled example kept together); stars from the same bootstrap. Absolute values are not comparable across rows because the validation slice moves with  $n$ ; its click/scroll/type counts are given per row. The study is descriptive: 24 contrasts, no multiplicity correction.

| $n_{\text{train}}$ | val mix (c/s/t) | A hit@0.10        | B - A                      | D-hook - A                 | B - A (hit@0.25)           | D-hook - A (hit@0.25)      |
|--------------------|-----------------|-------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| 300                | 170/33/47       | 0.139 $\pm$ 0.032 | +0.067 [+0.031, +0.103]*** | +0.069 [+0.041, +0.099]*** | -0.055 [-0.092, -0.019]**  | +0.067 [+0.033, +0.101]*** |
| 500                | 198/12/40       | 0.315 $\pm$ 0.059 | +0.011 [-0.019, +0.040]    | +0.001 [-0.032, +0.033]    | +0.004 [-0.027, +0.033]    | +0.059 [+0.025, +0.092]*** |
| 800                | 141/71/38       | 0.261 $\pm$ 0.020 | -0.040 [-0.073, -0.008]*   | +0.048 [+0.009, +0.087]*   | -0.039 [-0.069, -0.008]*   | +0.064 [+0.025, +0.103]*** |
| 1200               | 169/46/35       | 0.255 $\pm$ 0.026 | +0.051 [+0.021, +0.081]*** | +0.045 [+0.015, +0.079]**  | +0.071 [+0.036, +0.107]*** | +0.055 [+0.017, +0.092]**  |
| 2500               | 166/53/31       | 0.244 $\pm$ 0.037 | -0.012 [-0.039, +0.015]    | +0.037 [+0.009, +0.067]**  | -0.008 [-0.043, +0.028]    | +0.104 [+0.067, +0.143]*** |
| 5000               | 187/29/34       | 0.363 $\pm$ 0.009 | +0.005 [-0.025, +0.037]    | +0.043 [+0.011, +0.076]**  | -0.023 [-0.051, +0.005]    | +0.032 [+0.001, +0.064]*   |

Table 9: D-token causal-use test: the same trained model evaluated with the gold action embedding, a wrong (cyclically permuted) one, and a zeroed one (5 seeds, paired bootstrap over pooled examples).

| Conditioning | hit@0.05          | hit@0.10                   | hit@0.25                   | mean L2 $\downarrow$       |
|--------------|-------------------|----------------------------|----------------------------|----------------------------|
| gold         | 0.140 $\pm$ 0.059 | 0.236 $\pm$ 0.058          | 0.502 $\pm$ 0.045          | 0.386 $\pm$ 0.020          |
| wrong        | 0.014 $\pm$ 0.004 | 0.079 $\pm$ 0.013          | 0.200 $\pm$ 0.027          | 0.728 $\pm$ 0.081          |
| zero         | 0.074 $\pm$ 0.055 | 0.167 $\pm$ 0.066          | 0.416 $\pm$ 0.103          | 0.489 $\pm$ 0.059          |
| gold - wrong |                   | +0.157 [+0.114, +0.201]*** | +0.302 [+0.246, +0.356]*** | -0.342 [-0.375, -0.309]*** |
| gold - zero  |                   | +0.069 [+0.038, +0.100]*** | +0.086 [+0.056, +0.117]*** | -0.103 [-0.121, -0.086]*** |

543 the localization is not an artifact of the supplied x. The interventions move attention the same way  
544 they move accuracy on this seed (Figure 3, Table 10): a wrong type roughly halves the target share  
545 for D-hook, and zeroing costs D-token both attention and accuracy. On this one seed zeroing left  
546 D-hook’s attention and accuracy unchanged, which the five-seed study in Section 6.1 shows to be  
547 within the noise of a small real cost. Table 10 gives the target-attention share for the last prompt  
548 token (the one that predicts the opening parenthesis), the token that predicts the first x digit, and the  
549 token that predicts the first y digit.

Table 10: Share of image attention within 0.10 of the target (chance 0.026) at three probe positions, seed 42, 120 screens, 3/4-depth layer, gold answer teacher-forced.

| Model   | Position          | gold  | wrong (cyclic) | wrong (click $\leftrightarrow$ scroll) | zero  |
|---------|-------------------|-------|----------------|----------------------------------------|-------|
| D-hook  | last prompt token | 0.040 | 0.036          | 0.035                                  | 0.036 |
| D-hook  | pre-x token       | 0.039 | 0.036          | 0.037                                  | 0.039 |
| D-hook  | pre-y token       | 0.111 | 0.064          | 0.049                                  | 0.119 |
| D-token | last prompt token | 0.038 | 0.034          | 0.034                                  | 0.035 |
| D-token | pre-x token       | 0.038 | 0.034          | 0.033                                  | 0.035 |
| D-token | pre-y token       | 0.095 | 0.080          | 0.079                                  | 0.079 |

## 550 H Qualitative examples

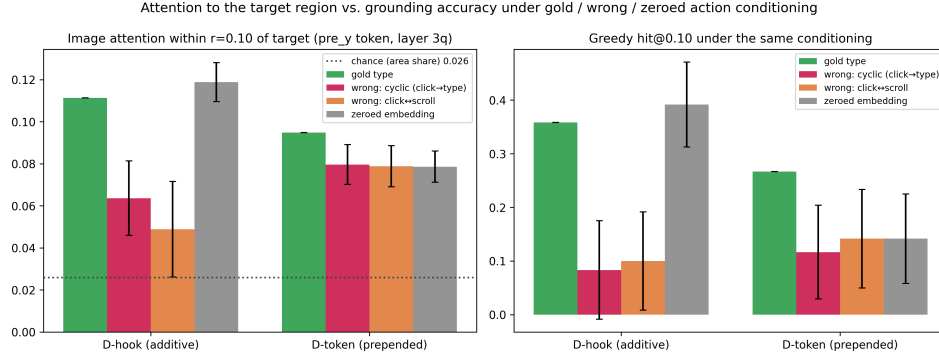


Figure 3: Same trained model, four conditionings, 120 validation screens, one seed per variant. Left: share of image attention within 0.10 of the target from the token that predicts the y coordinate, with the gold answer teacher-forced (3/4-depth layer, head-averaged); the dotted line is the share of image tokens within that radius. Right: greedy hit@0.10 under the same conditioning. Error bars are 95% paired-bootstrap intervals of the difference from gold over the 120 screens.

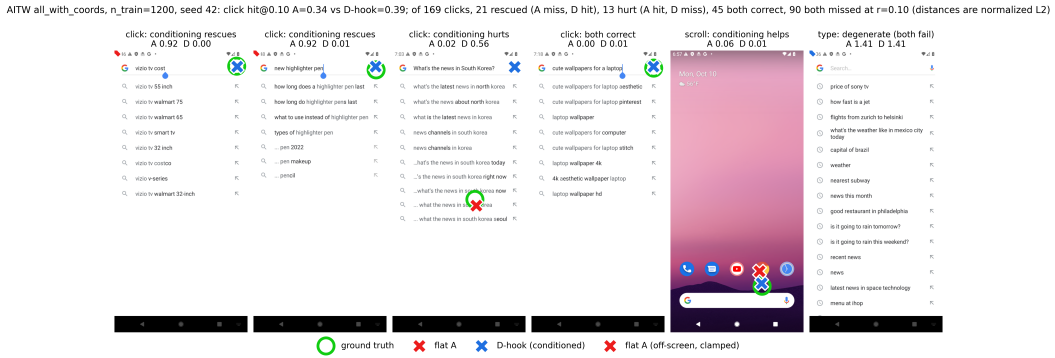


Figure 4: Predictions of the flat baseline (red) and D-hook (blue) against the target (green) on held-out AITW screens at the headline setting, one seed. The panels are a fixed spread of outcomes rather than a selection of wins: two rescues, a case where conditioning hurts, a case where both are right, a scroll, and a degenerate *type* example on which both models miss at distance  $\sqrt{2}$ . The title reports the base rates on this seed at  $r = 0.10$ : of 169 clicks, 21 are rescued by conditioning, 13 are hurt, 45 are correct under both models, and 90 are missed by both. In the rescue panels the baseline's prediction is off-screen and drawn clamped at the corner.